

S.E. (Computer / Information Technology)
Engineering Mathematics - III
2012 Course

Time: 2Hours

Max. Marks : 50

Instructions to the candidates:

- 1) Solve Q1 or Q2, Q3 or Q4, Q5 or Q6 and Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Use of Calculator is allowed.
- 5) Assume Suitable data if necessary

Q1) a) Solve any **two** of the following [08]

i) $(D^2 + 4)y = \cos 3x \cdot \cos x$

ii) $(D^2 + 6D + 9)y = e^{3x} / x^2 \dots$ (by variation of parameters method)

iii) $x^3 (d^3y / dx^3) + 2x^2 (d^2y / dx^2) + 2y = 20 (x + 1/x)$

b) Obtain f(k) given that, [04]

$$f(k+2) + 5f(k+1) + 6f(k) = 0, \quad k \geq 0 \quad f(0) = 0, \quad f(1) = 2$$

by using Z transform.

OR

Q2) a) An emf $E \sin(pt)$ is applied at $t = 0$ to a circuit containing a condenser 'C' and [04]

Inductance 'L' in series. The current 'x' satisfies the equation

$$L(dx/dt) + \frac{1}{C} \int x dt = E \sin(pt)$$

Where $= \frac{-dq}{dt}$. If $p^2 = \frac{1}{LC}$ and initially the current x and charge q is zero then show that current in the circuit at any time t is $\frac{E}{2L} t \sin(pt)$.

b) Solve the integral equation [04]

$$\int_0^\infty f(x) \cos \lambda x dx = \begin{cases} 1 - \lambda & 0 \leq \lambda \leq 1 \\ 0 & \lambda > 1 \end{cases}$$

And hence show that $\int_0^\infty \frac{\sin^2 z}{z^2} dz = \pi/2$

c) Attempt **any one** [04]

i) Find the Z- transform of $f(k) = e^{-2k} \cos(5k + 3)$

ii) Find the inverse Z- transform of $\frac{z(z+1)}{z^2-2z+1}$, $|z| > 1$

Q3) a) The first four moments of a distribution about 2 are 1, 2.5, 5.5 and 16. Calculate the first four moments about the mean, A. M., S. D., β_1 and β_2 [05]

b) In a certain examination 200 students appeared. Average marks obtained were 50% with standard deviation 5%. How many students do you expect to obtain more than 60% of marks, supposing that the marks are distributed normally ? [04]
(Given $z=2$; $A = 0.4772$)

c) Find the directional derivatives of: [04]
 $\phi = xy^2 + yz^2 + zx^2$ at (1,1,1) along the line $2(x-2) = y+1 = z-1$

OR

Q4) a) Calculate the coefficient of correlation for the following data [05]

x	1	2	3	4	5	6	7	8	9
y	9	8	10	12	11	13	14	16	15

b) Prove the following (**Any one**) [04]

i) $\nabla^4 r^4 = 120$

ii) $\nabla \cdot \left[r \nabla \frac{1}{r^5} \right] = \frac{15}{r^6}$

c) Show that $\vec{F} = (x^2 - yz)\vec{i} + (y^2 - zx)\vec{j} + (z^2 - xy)\vec{k}$ is irrotational. Also find Φ such that $\vec{F} = \nabla\Phi$ [04]

Q5) a) Find the work done in moving a particle along [04]

$x=a \cos\theta, y=a \sin\theta, z=b\theta$, from $\theta = \frac{\pi}{4}$ to $\theta = \frac{\pi}{2}$ under a field of force given

by $\vec{F} = -3a \sin^2\theta \cos\theta \vec{i} + a(2\sin\theta - 3\sin^3\theta)\vec{j} + b \sin 2\theta \vec{k}$

b) Evaluate $\iint_s (yz\vec{i} + zx\vec{j} + xy\vec{k}) \cdot d\vec{s}$, where s is the curved surface of the cone $x^2 + y^2 = z^2, z = 4$ [04]

c) Using Stokes Theorem to evaluate $\int_C (4y\vec{i} + 2z\vec{j} + 6y\vec{k}) \cdot d\vec{r}$ where C is the curve of intersection of $x^2 + y^2 + z^2 = 2z$ and $x = z - 1$ [04]

OR

Q6) a) A vector field is given by $\vec{F} = (2x - \cos y)\vec{i} + x(4 + \sin y)\vec{j}$, evaluate $\int_C \vec{F} \cdot d\vec{r}$, [04]

where c is the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, z = 0$.

- b) Prove that $\iiint_v \frac{1}{r^2} dv = \iint_s \frac{1}{r^2} \vec{r} \cdot d\vec{s}$, where s is closed surface enclosing the volume v . Hence evaluate $\iint_s \frac{x\hat{i}+y\hat{j}+z\hat{k}}{r^2} \cdot d\vec{s}$, where s is the surface of the sphere $x^2+y^2+z^2 = a^2$. [04]

- c) If $\vec{E} = \nabla\phi$, and $\nabla^2\phi = -4\pi\rho$, [04]
 prove that $\iint_s \vec{E} \cdot d\vec{s} = -4\pi \iiint_v \rho dv$

- Q7) a) Find the value of p such that the function [04]
 $f(x) = r^2 \cos 2\Theta + i r^2 \sin p\Theta$ becomes analytical function.

- b) Evaluate $\oint_c \frac{z^2 + \cos^2 z}{(z - \frac{\pi}{4})^3} dz$ [05]

where c is a circle $x^2+y^2=1$

- c) Find the bilinear transformation which maps $1, i, -1$ from z plane into $i, 0, -i$ from the w plane [04]

OR

- Q8) a) Determine the analytic function $f(z)$ whose real part is [04]

$$U = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$$

- b) Evaluate $\oint_c \left[\frac{\sin \pi z^2 + 2z}{(z-1)^2(z-2)} \right] dz$ [05]

where c is a circle $x^2+y^2=16$

- c) Show that the transformation $\omega = \sin z$ transforms the straight lines $x = c$ of z plane into hyperbolas in the ω plane [04]

***** E N D *****